CHEMICAL ENGINEERING · MACHINE LEARNING R&D

AI-Based Crude Oil Characterization and Refinery Yield Prediction

An ML-Based Decision-Support Approach for Rapid Crude Feedstock Assessment

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Course: CH330- Seminar

INTRODUCTION

Why Crude-Oil Variability Matters to a Refinery

Crude oils are not chemically or physically identical. Properties such as API/specific gravity, sulfur content, nitrogen content, viscosity, and pour point vary substantially between sources — and these differences strongly influence which refining processes a crude is suited for and what fractions it can yield.

9,056

historical crude-oil analyses in the COA Database
(8,500 U.S. domestic oils) — DOE / Bartlesville Project Office



This chain is what our project asks a model to learn directly from historical laboratory data.

MEASURED FOR MOST SAMPLES IN THE DATABASE

- API / Specific Gravity
- Sulfur Content
- Nitrogen Content
- Viscosity (77°F & 100°F)
- Pour Point
- Color

Evaluated as candidate model inputs, per the Bureau of Mines routine analysis procedure in the COA codebook.

The COA Database contains:

- Physical/chemical properties: gravity, sulfur, nitrogen, viscosity, pour point
- Geological & geographical information
- Distillation fraction volume percentages
- Properties of distillation fractions & residuum

KNOWLEDGE GAP & IMPORTANCE

From Days of Lab Work to a Rapid Data-Driven Estimate

TRADITIONAL WORKFLOW

Crude sample

↓

Laboratory characterization

↓

Distillation / analysis

↓

Interpretation

↓

Engineering decision

PROPOSED WORKFLOW

Crude properties

↓

ML model

↓

Rapid predicted fraction profile

↓

Preliminary decision support

↓

Detailed analysis if required

The Knowledge Gap

Can historical crude-oil characterization data be used to learn the relationship between readily measured crude properties and its refining-relevant distillation profile?

Potential Industry Value

- Faster preliminary screening of crude feedstocks
- Data-driven feedstock assessment
- Potential reduction in repetitive preliminary analysis
- Early identification of crude characteristics
- Support for refinery planning
- Better utilization of decades of historical crude-oil data

Note: outcomes are described as potential and preliminary — this tool is not positioned to guarantee cost savings or replace laboratory testing.

OBJECTIVES

Project Objectives

1. Preprocess and clean the COA crude-oil dataset
2. Identify meaningful crude-oil properties through exploratory and correlation analysis
3. Reduce redundant variables and select an informative feature subset
4. Develop ML models for predicting selected crude-oil distillation/yield characteristics
5. Compare Linear Regression and Random Forest Regression using R^2, MAE, and RMSE
6. Identify the better-performing model for the intended prediction task
7. Develop a simple software interface for rapid crude-feed assessment
8. Demonstrate ML's potential as a decision-support tool for preliminary refinery screening

WORK FLOW · PART 1 OF 2

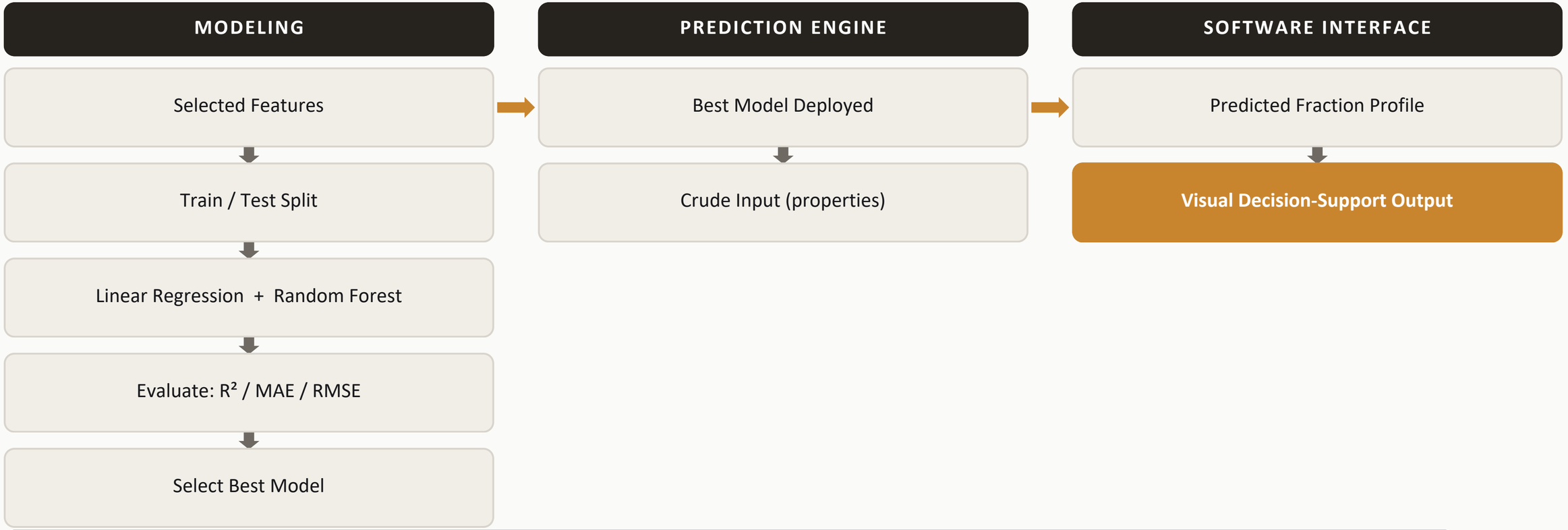
System Architecture: From Raw Data to Final Dataset



Note: unnecessary identification, geographical, and administrative code fields (e.g. county codes, sample IDs) will not automatically be used as ML input features. Only chemically and physically meaningful variables are retained.

https://catalog.data.gov/dataset/crude-oil-analysis-coa-database?

System Architecture: Modeling to Software Interface



Streamlit-style software interface: engineer enters selected crude properties → model returns predicted fraction characteristics → results are visualized (e.g., bar chart of predicted fractions).

Methodology: Data & Feature Engineering

1 Understand the Dataset

Study variable definitions using the official COA documentation and codebook.

2 Clean the Data

Handle missing, invalid, and inconsistently coded values across the dataset.

3 Separate Variable Types

Identification vs. geological/geographical vs. physical/chemical vs. distillation/yield variables.

4 Correlation Analysis

Quantify relationships between candidate input properties and target fractions.

5 Identify Redundant Features

Flag highly correlated or duplicated information for removal.

6 Select Final Feature Set

Retain a compact, interpretable, chemically meaningful set of inputs.

“We do not intend to feed all available variables into the model — the objective is a compact, interpretable, and chemically meaningful feature set.”

Methodology: Modeling & Software Interface

Linear Regression — Baseline

Used as an interpretable baseline to determine how well relatively simple, linear relationships can predict the crude's distillation/yield characteristics.

Random Forest Regression — Main Model

Used to capture nonlinear relationships between crude properties and predicted distillation/yield outputs.

Evaluation Metrics

R^2 | MAE | RMSE

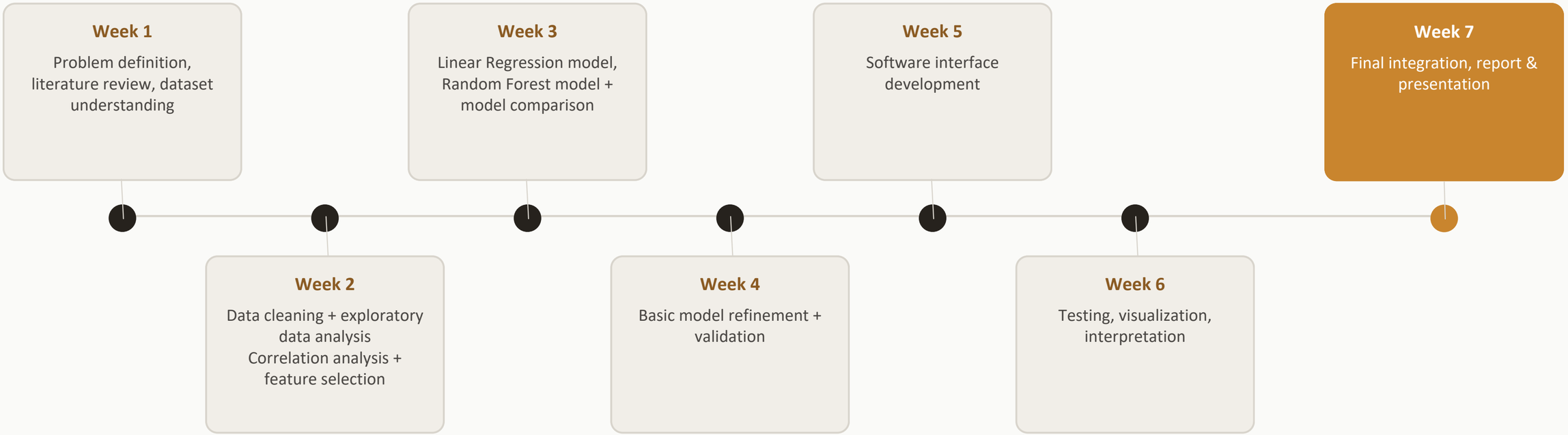
Metrics will be computed after training; no results exist yet at the proposal stage.

Software Interface Flow

User enters crude properties → Model predicts fraction profile → Results shown as visual chart → **Rapid preliminary crude assessment**

TIMELINE

Proposed 7-Week Work Plan



A 2–3 month course project; timeline can compress to 8 weeks if required.

Thank You

“Turning crude-oil data into refinery intelligence.”

